

Simultaneous equations and inequalities - Set 1

TMUA practice questions by topic. Specification MM1.4, MM1.5, M4.16, M4.17. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 Find the complete set of values of x for which $2x^2 - 5x - 3 \geq 0$.
- A** $x \leq -\frac{1}{2}$ or $x \geq 3$
- B** $x \leq -\frac{1}{2}$
- C** $x \geq 3$
- D** $x \leq -3$ or $x \geq \frac{1}{2}$
- E** $-3 \leq x \leq \frac{1}{2}$
- F** $-\frac{1}{2} \leq x \leq 3$
- 2 Find the set of values of x satisfying both $3x - 7 < 2$ and $x^2 > 4$.
- A** $x < -2$ or $2 < x < 3$
- B** $2 < x < 3$
- C** $-2 < x < 3$
- D** $x < 3$
- E** $x > 2$
- F** $x < -2$
- 3 The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point. Find the possible values of k .
- A** $k = \pm 2\sqrt{2}$
- B** $k = \pm 8$
- C** $k = \pm 4$
- D** $k = \pm \sqrt{8}$
- E** $k = 4$ only
- F** $k = \pm 2$

- 4 For how many integers n is $n^2 - 10n + 16 < 0$?
- A 5
B 3
C 4
D 8
E 7
F 6
- 5 Solve $xy = 6$ and $x - y = 1$. What is the sum of all possible values of $x + y$?
- A 10
B 5
C -5
D 7
E 1
F 0
- 6 The inequality $x^2 + bx + 9 > 0$ holds for all real x . Find the complete set of values of b .
- A $-3 < b < 3$
B $b < -6$ or $b > 6$
C $b > 6$
D $-6 < b < 6$
E $b \geq 6$
F all real b
- 7 The simultaneous equations $y = x^2 + 2x + c$ and $y = 4x + 1$ have exactly one solution. Find the value of c .
- A 3
B -1
C 2
D 1
E -2
F 0

- 8** The curves $y = x^2 - 4$ and $y = -x^2 + 2x + k$ intersect at exactly two points. Find the complete set of values of k .
- A** $k > \frac{9}{2}$
- B** $k < -\frac{9}{2}$
- C** $k > -\frac{9}{2}$
- D** $k \neq -\frac{9}{2}$
- E** $k > -4$
- F** all real k
- 9** Solve $2x + 3y = 12$ and $x^2 - y^2 = 0$. How many solutions (x, y) are there?
- A** 0
- B** 2
- C** 1
- D** 4
- E** 3
- F** infinitely many
- 10** Two numbers have sum 10 and the sum of their squares is 58. What is the positive difference between them?
- A** 3
- B** 4
- C** 6
- D** 1
- E** 2
- F** 8

- 11** x is a real number. P : ' $x^2 < 4$ '. Q : ' $x < 2$ '. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .
- 12** Solve the simultaneous equations $x + 2y = 7$ and $x^2 + y^2 = 10$. Find the sum of the possible values of x .
- A** 7
 - B** 2
 - C** $\frac{7}{5}$
 - D** $\frac{14}{5}$
 - E** $\frac{28}{5}$
 - F** 4